Monitoring Report

Qingdao Longxin Wind Power Project Phase I

Document Prepared By Goldchina Consultancy International Co., Ltd.

Project Title	Qingdao Longxin Wind Power Project Phase I
Version	02
Report ID	-
Date of Issue	30-01-2013
Project ID	CDM registered reference number: 6697
Monitoring Period	08-11-2011 to 27-09-2012 (including both days)
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1 PROJECT DETAILS

1.1 Summary Description of Project

Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the Project) developed by China Resources Wind Power (Qingdao) Co., Ltd., involves construction and operation of a wind power project that is sited on Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The Project was commenced on 08/11/2011. The Project has been registered as a CDM project on 28/09/2012[1] (UNFCCC registration reference number: 6697).

The purpose of the Project is to install 24 wind turbines with capacity of 2000KW each and 1 wind turbine with capacity of 1800kW to generate clean renewable electricity. The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing part of electricity generated by NCPG which is dominated by fossil fuel fired power plants.

Wind turbines and their corresponding generators employed by the Project were provided by Vestas Wind Technology (China) Co., Ltd.

Relevant dates for the project activity is as below:

Construction start date: 26/05/2011

Commissioning start date: 08/11/2011

Continued operation periods: 20 years

During the monitoring period (08/11/2011 - 27/09/2012, including both days), the monitoring activities were conducted strictly in accordance with the monitoring plan in the registered PDD. The Project has operated without any accidental or emergency events that might impact the accuracy and/or implementation of monitoring activities. The net feed-in electricity during this period is 88,417.56MWh. The total emission reductions in this monitoring period are 82,302tCO_2e.

1.2 Sectoral Scope and Project Type

This category would fall within sectoral scope 1: energy industries (Renewable sources).

The type of the Project is a wind power generation project.

1.3 Project Proponent

Organization:	China Resources Wind Power (Qingdao) Co., Ltd.
Street/P.O.Box:	No.5002, Shennan Dong Road Luohu District
Building:	Floor 28, Di Wang Building
City:	Shenzhen
State/Region:	-
Postfix/ZIP:	518001

[1] http://cdm.unfccc.int/Projects/DB/TUEV-RHEIN1342414627.77/view

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1.4 Other Entities Involved in the Project

There is no any other project participant(s).

1.5 Project Start Date

Commissioning start date: 08/11/2011

1.6 Project Crediting Period

Project start date: 08/11/2011 on which the project activity began reducing or removing GHG emission. And the Project has been registered successfully as a CDM project on 28/09/2012. That is, the period from 08/11/2011 to 27/09/2012 is considered as the Project crediting period.

1.7 Project Location

The Project is located at Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The nearest Liyuan Town is about 10 kilometers from the project site. The geographical coordinates of the project is from 119.8827°E~119.9503°E and from 36.8157°N~37.8915°N (The central GPS coordinate is 119.91125°E, 36.8568°N).

According to the methodology ACM0002, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to.

1.8 Title and Reference of Methodology

Approved consolidated baseline and monitoring methodology ACM0002 (version 12.2.0)-*Consolidated methodology for grid-connected electricity generation from renewable sources*.

2 IMPLEMENTATION STATUS

2.1 Implementation Status of the Project Activity

The project started operation on 08/11/2011.

Bidirectional meter M1 with accuracy 0.5S was installed on the Tangtian Substation as the main meter to measure the electricity supplied by the Project and Phase II project, meanwhile M1 is used to measure the electricity imported from the grid by the Project and Phase II project. Meter M2 and M3 installed at the on-site substation recorded the electricity supplied by the Project and Phase II Project respectively.
Meter M4 owned by the project owner is used to measure the electricity imported from the grid by the project and Phase II project via spare 10kV back-up line.

The Project had been in normal operation during the monitoring period, without special events such as overhaul or exchange of equipment. There were no events or situations which may impact the applicability of the methodology.
The monitoring system was implemented in line with the monitoring plan in the registered PDD. The actual operation was conducted strictly in compliance with the description in the monitoring plan.
During the monitoring period, there were no other special events including malfunction and emergencies affected the applicability of methodology occurred.

2.2 Project Description Deviations

There is no deviation applied to this monitoring period.

2.3 Grouped Project

The Project is not a grouped project.

3 DATA AND PARAMETERS

3.1 Data and Parameters Available at Validation

Data Unit / Parameter:	$EF_{grid,CM,y}$
Data unit:	tCO_2e/MWh
Description:	Baseline emission factor of NCPG in the monitoring period.
Source of data:	Notification on 2010 baseline emission factors for regional power grids in China, issued by China on Dec 20, 2010 and the registered PDD.
Value applied:	0.9309
Purpose of the data:	This data is used for baseline emission calculations

Any comment:	This parameter is ex ante determined in PDD and fixed during the first crediting period.

3.2 Data and Parameters Monitored

Data Unit / Parameter:	$EG_{output,y}$
Data unit:	MWh/yr
Description:	Total electricity supplied to the Grid by the Project and Phase II project in the monitoring period
Source of data:	Measured by the electricity meter M1
Description of measurement methods and procedures to be applied:	The readings of the electricity meter were continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording:	Continuously measured and monthly recorded
Value monitored:	120,645.36
Monitoring equipment:	M1 is a bidirectional meter. Type: DTZ341 Accuracy class: 0.5S Serial number: QD0243006231 Calibration frequency: at least annually Calibration date: 20/09/2011, 18/09/2012. Calibration validity: till 17/09/2013
QA/QC procedures to be applied:	The accuracy of meters is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000).
Calculation method:	NA
Any comment:	

Data Unit / Parameter:	$EG_{another,y}$
Data unit:	MWh/yr
Description:	Electricity generated by the Phase II project in the monitoring period

Source of data:	Measured by the electricity meter M3
Description of measurement methods and procedures to be applied:	The readings of the electricity meter were continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording:	Continuously measured and monthly recorded
Value monitored:	31,806.72
Monitoring equipment:	M3 is a directional meter. Type: DTSD341 Accuracy class: 0.5S Serial number: 11060727990140 Calibration frequency: at least annually Calibration date: 10/04/2012. Calibration validity: till 09/04/2013
QA/QC procedures to be applied:	The accuracy of meters is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000).
Calculation method:	NA
Any comment:	

Data Unit / Parameter:	$EG_{import,y}$
Data unit:	MWh/yr
Description:	Total electricity imported from the Grid by the project and Phase II project through the 110KV line in the monitoring period
Source of data:	Measured by the electricity meter M1
Description of measurement methods and procedures to be applied:	The readings of the electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording:	Continuously measured and monthly recorded
Value monitored:	421.08
Monitoring equipment:	M1 is a bidirectional meter. Type: DTZ341 Accuracy class: 0.5S Serial number: QD0243006231

	Calibration frequency: at least annually Calibration date: 20/09/2011, 18/09/2012. Calibration validity: till 17/09/2013
QA/QC procedures to be applied:	The accuracy of meters is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000).
Calculation method:	NA
Any comment:	

Data Unit / Parameter:	$EG_{im\text{-}spare,y}$
Data unit:	MWh/yr
Description:	Electricity imported from the Grid by the project and Phase II Project through a spare 10kV line in the monitoring period
Source of data:	Measured by the electricity meter M4
Description of measurement methods and procedures to be applied:	The local power distributor measured the $EG_{im\text{-}spare,y}$ by metering instrument operated by the Grid Company.
Frequency of monitoring/recording:	Continuously measured and monthly recorded
Value monitored:	0
Monitoring equipment:	M4 is a unidirectional meter. Type: DTZ522 Accuracy class: 1.0 Serial number: DT4803509 Calibration frequency: at least annually Calibration date: 14/10/2011. Calibration validity: till 13/10/2012
QA/QC procedures to be applied:	The accuracy of meters is 1.0, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000).
Calculation method:	NA
Any comment:	

Data Unit / Parameter:	$EG_{facility,y}$
Data unit:	MWh/yr
Description:	Net electricity supplied to the Grid by the proposed project in the year y.
Source of data:	Measured by the electricity meter M1, M3 and M4.
Description of measurement methods and procedures to be applied:	The readings of M1, M3 and M4 were continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording:	Continuously measured and monthly recorded
Value monitored:	88,417.56
Monitoring equipment:	M1 is a bidirectional meter. Type: DTZ341 Accuracy class: 0.5S Serial number: QD0243006231 Calibration frequency: at least annually Calibration date: 20/09/2011, 18/09/2012 Calibration validity: till 17/09/2013 M3 is a directional meter. Type: DTSD341 Accuracy class: 0.5S Serial number: 11060727990140 Calibration frequency: at least annually Calibration date: 10/04/2012. Calibration validity: till 09/04/2013 M4 is a unidirectional meter. Type: DTZ522 Accuracy class: 1.0 Serial number: DT4803509 Calibration frequency: at least annually Calibration date: 14/10/2011. Calibration validity: till 13/10/2012
QA/QC procedures to be applied:	The accuracy of M1, M3 and M4 is 0.5S, 0.5S and 1.0 respectively, which is consistent with Technical administrative code of electric energy metering (DL/T448-2000). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000).
Calculation method:	The net electricity supplied to the Grid by the proposed project were calculated by the following

	formula: $EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im\text{-}spare,y}$
Any comment:	

3.3 Description of the Monitoring Plan

1. Data collection procedures
   The electric energy metering system was equipped according to the requirements of the *Technical administrative code of electric energy metering* (DL/T448-2000).

   The electricity supplied by the Project to the Grid will be transmitted from the on-site substation of the Project through the 110kV transmission line and then be delivered to the Grid and share the 110kV transmission line with the Phase II Project, which is developed by the same project owner with the Project.

   Bidirectional meter M1 installed on the Tangtian Substation as the main meter is used to measure the electricity ($EG_{output,y}$) supplied by the Project and Phase II project. Meter M2 and M3 installed at the on-site substation are used to measure the electricity supplied by the Project and the electricity ($EG_{another,y}$) supplied by Phase II Project respectively.

   The Phase II Project started power generation and was combined to the grid on 16/05/2012, sharing the 110kV transmission line with the Phase I Project. After that, magnification of M1 became 132000 from 66000.

   The electricity ($EG_{import,y}$) that bought from the grid is measured by bidirectional meter M1 installed in Tangtian substation.

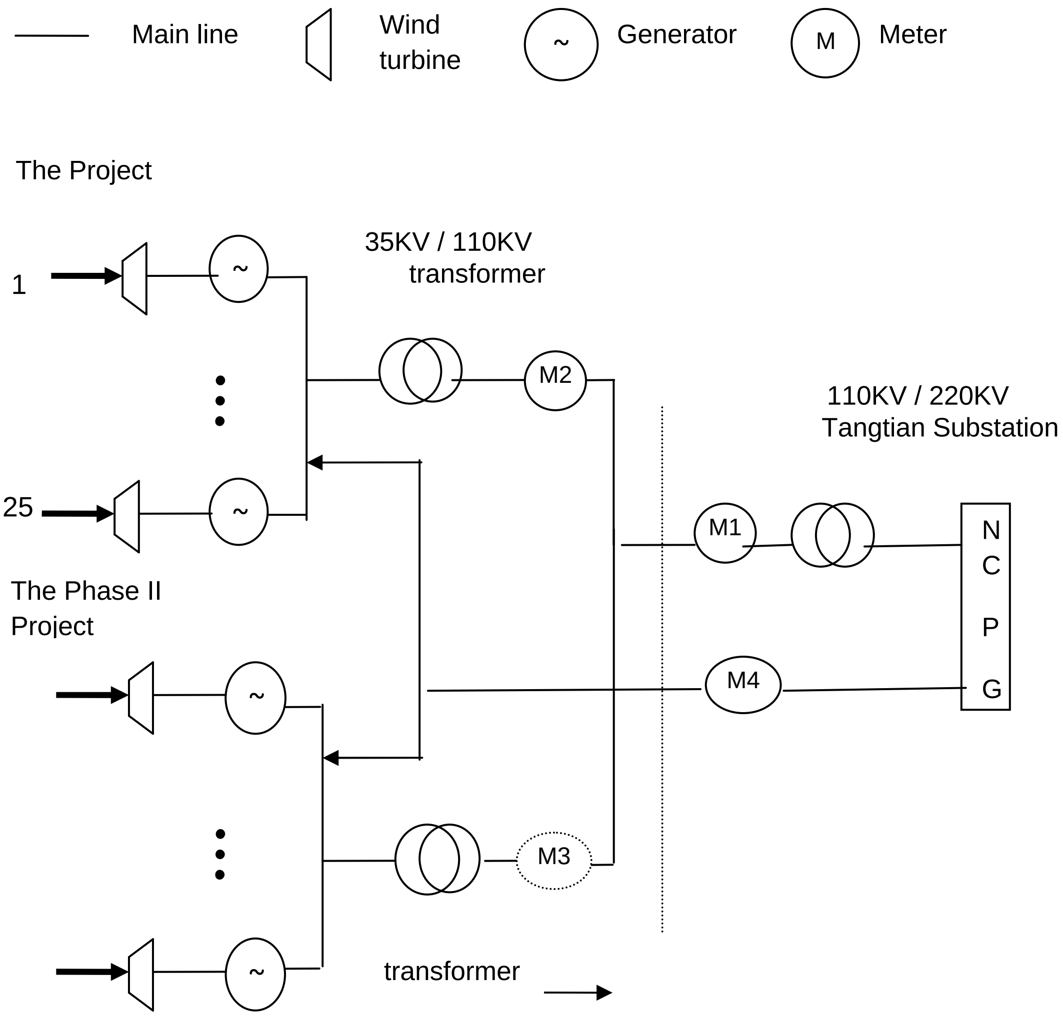

Figure 1: The process flowchart of the Project

Additionally, a spare 10kV line is used for supplying electricity in order to start-up or emergencies. Electricity ($EG_{im\text{-}spare,y}$) imported from the grid by the project via this spare 10kV line can be measured by a meter (M4) which is owned by the project owner.

The net electricity supplied to the grid by the Project is calculated as followed:

$$EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im\text{-}spare,y}$$

The reading of meter (M2) is carried out by the project owner for cross check.

The project owner keeps and safe keeps the records of the main meter's data readings for verification by the DOE.

Please see Figure 1 for more details.

2. Organizational structure

China Resources Wind Power (Qingdao) Co., Ltd., the project owner will use this document as guideline in monitoring of the project emission reduction performance and will adhere to the guidelines set out in this monitoring plan. This plan should be modified according to actual conditions and requirements of DOE in order to ensure that the monitoring is credible, transparent, and conservative.

The responsibilities of the project staff are as follow:

The General Manager of the project entity will appoint a CDM project manager. The Operational and Monitoring Manager of the plant, the Financial Chief, and the Technical Chief are responsible for the collection of the data and information required in the monitoring plan.
General Manager: To be responsible for supervising the whole monitoring procedure. CDM Project Manager: To be responsible for data management and compiling monitoring report. Operational and monitoring manager: To be responsible for collecting data and do internal audit.
Financial chief: To be responsible for collection of sales receipts.
Technical chief: To be responsible for preparing operational reports of the project activity, recording the daily operation of the wind farm, including operating periods, equipment defects, etc.

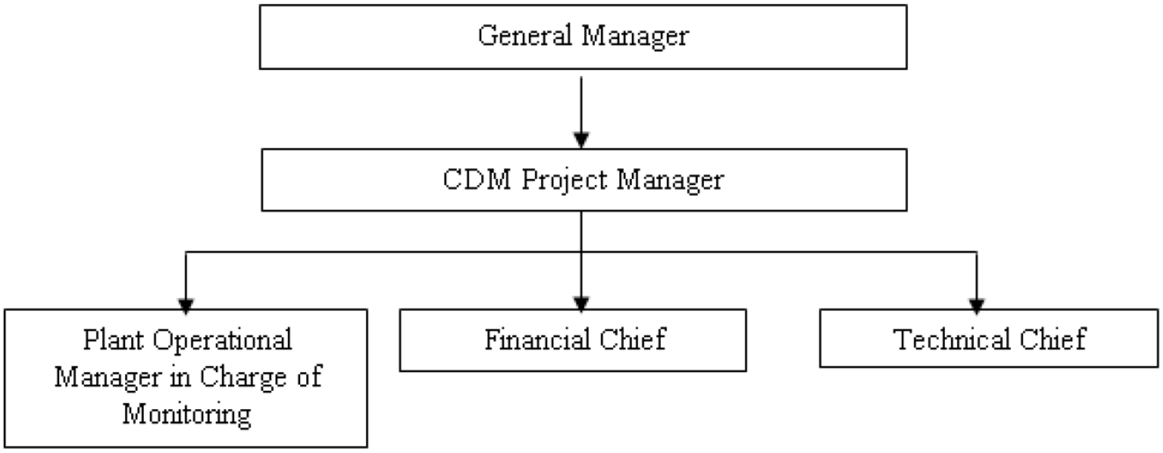

Figure 2. Management Structure of Monitoring Plan

3. Emergency procedures

In case metering equipment is damaged and no reliable readings can be recorded the project entity will estimate net supply by the proposed project activity according to the following procedure:

In case metering equipment is damaged: The project entity and the grid company will jointly calculate a conservative estimate of power supplied to the grid. A statement will be prepared indicating

I: the background to the damage to metering equipment;

II: the assumptions used to estimate net supply to the grid for the days for which no record could be recorded;

III: the estimation of power supplied to the grid. The statement will be signed by both a representative of the project entity as well as a representative of the grid company.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions in the monitoring period are calculated by multiplying the fixed baseline emission factor determined in the registered PDD of the project ex ante by net electricity supplied by the Project to the NCPG in the monitoring period.

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (1)$$

$EG_{PJ,y} = EG_{facility,y}$ (2)

$EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im\text{-}spare,y}$ (3)

Where:
BEy = Baseline emissions in year y. (tCO_2 /yr)
$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$ = Combined margin CO_2 emission factor for the project electricity system in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system".
$EG_{facility,y}$=Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)
$EG_{output,y}$: Total electricity supplied to the Grid by the proposed project and Phase II project in the year y, which will be measured by bidirectional meter M1 installed in the Tangtian substation;
$EG_{another,y}$: Electricity supplied to the Grid by the Phase II Project in the year y, which will be measured by electricity meter M3;
$EG_{import,y}$: Total electricity imported from the Grid by the proposed project and Phase II project in the year y, which will be measured by bidirectional meter M1 installed in Tangtian substation;
$EG_{im\text{-}spare,y}$: Electricity imported from the Grid through a spare 10 kV line in the year y, which will be measured by meter M4 installed on the spare power line on the site of the proposed project

The $EG_{facility,y}$ and BE_y are listed as follow:

Monitoring period	$EG_{output,y}$ (MWh/yr)	$EG_{another,y}$ (MWh/yr)	$EG_{import,y}$ (MWh/yr)	$EG_{im\text{-}spare,y}$ (MWh/yr)	$EG_{facility,y}$ (MWh/yr)	$EF_{grid,CM,y}$ (tCO2e/MWh)	BE_y
	A	B	C	D	E=A-B-C-D	F	G=E×F
08/11/2011-25/11/2011	1,943.70	0	19.8	0	1,923.90	0.9309	1,790
26/11/2011-25/12/2011	6,586.14	0	13.2	0	6,572.94	0.9309	6,118
26/12/2011-31/12/2011	662.64	0	15.84	0	646.8	0.9309	602
01/01/2012-25/01/2012	4,787.64	0	67.32	0	4,720.32	0.9309	4,394
26/01/2012-25/02/2012	12,045.00	0	23.76	0	12,021.24	0.9309	11,190

26/02/2012-25/03/2012	10,684.08	0	28.38	0	10,655.70	0.9309	9,919
26/03/2012-25/04/2012	19,763.70	0	8.58	0	19,755.12	0.9309	18,390
26/04/2012-25/05/2012	11,571.78	2,256.32	27.72	0	9,287.74	0.9309	8,645
26/05/2012-25/06/2012	16,785.12	9,759.20	31.68	0	6,994.24	0.9309	6,510
26/06/2012-25/07/2012	12,175.68	6,839.36	43.56	0	5,292.76	0.9309	4,927
26/07/2012-25/08/2012	11,824.56	6,469.76	72.6	0	5,282.20	0.9309	4,917
26/08/2012-27/09/2012	11,815.32	6,482.08	68.64	0	5,264.60	0.9309	4,900
Total	120,645.36	31,806.72	421.08	0	88,417.56		82,302

BE_y=82,302tCO_2e

4.2 Project Emissions

The Project is a zero-emission renewable electricity generating activity. Therefore no emission from the project activity was identified, and $PE_y = 0$.

4.3 Leakage

According to the methodology ACM0002, the project leakage is not considered, and therefore, L_y =0.

4.4 Summary of GHG Emission Reductions and Removals

$$ER_y = BE_y - PE_y$$

Total baseline emissions in the monitoring period: BE_y =82,302 tCO_2e

Total project emissions in the monitoring period: $PE_y = 0$

Total emission reductions in the monitoring period: ER_y =82,302 tCO_2e

The monthly GHG Emission Reductions are shown as follow:

period	Baseline emission	Project activity emission	Leakage	emission reductions
	BEy	PEy	Ly	ERy
	tCO_2e	tCO_2e	tCO_2e	tCO_2e
08/11/2011-25/11/2011	1,790	0	0	1,790
26/11/2011-25/12/2011	6,118	0	0	6,118
26/12/2011-31/12/2011	602	0	0	602
Total for 2011	8,510	0	0	8,510
01/01/2012-25/01/2012	4,394	0	0	4,394
26/01/2012-25/02/2012	11,190	0	0	11,190
26/02/2012-25/03/2012	9,919	0	0	9,919
26/03/2012-25/04/2012	18,390	0	0	18,390
26/04/2012-25/05/2012	8,645	0	0	8,645
26/05/2012-25/06/2012	6,510	0	0	6,510
26/06/2012-25/07/2012	4,927	0	0	4,927
26/07/2012-25/08/2012	4,917	0	0	4,917
26/08/2012-27/09/2012	4,900	0	0	4,900
Total for 2012	73,792	0	0	73,792
Total	82,302	0	0	82,302

The emission reductions for 2011 are 8,510 tCO_2e, for 2012 are 73,792 tCO_2e.

5 ADDITIONAL INFORMATION

During the monitoring period (08/11/2011to 27/09/2012), the emission reductions are 82,302tCO_2e, about 0.27% higher than the ex-ante estimated emission reductions of the registered CDM-PDD, and the increase is slight and reasonable.